

PROJECT 22

Combustion Chamber — Haiwell SCADA Interface

Effective Engineering (Client: University of Moratuwa)
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PROJECT SUMMARY

Contributed to the SCADA monitoring system for a combustion chamber research rig at the University of Moratuwa. Used Haiwell SCADA software to build and enhance the GUI, creating dynamic displays for temperature, pressure, and flow rate that the research team could read in real time.

OBJECTIVES

- Set up real-time monitoring of combustion chamber parameters (temperature, pressure, flow).
- Enhance the Haiwell SCADA GUI for better operator usability.
- Create clear dynamic displays for process variables.

HOW IT WORKS

Haiwell SCADA connected to Haiwell PLC which reads sensors on the combustion chamber rig. SCADA displays process parameters live on a custom GUI with trend graphs, numeric readouts, and alarm indicators. The GUI was customised to the research team's specific monitoring needs.

KEY COMPONENTS

Component	Specification / Function
Haiwell SCADA Software	Supervisory control and data display
Haiwell PLC	Sensor data acquisition
Temperature / Pressure / Flow Sensors	Process instrumentation
Custom GUI Screens	Operator and researcher interface

TECHNICAL SPECIFICATIONS

Parameter	Value	Notes
SCADA platform	Haiwell SCADA	—
Parameters monitored	Temperature, pressure, flow rate	Real-time
[TO BE FILLED]	Number of SCADA screens	—

MY CONTRIBUTION

- Modified and enhanced the Haiwell SCADA GUI for the research team.
- Designed dynamic displays for temperature, pressure, and flow rate.
- Supported live test sessions and data verification.
- [TO BE FILLED] — any alarm or data logging work.

CHALLENGES & SOLUTIONS

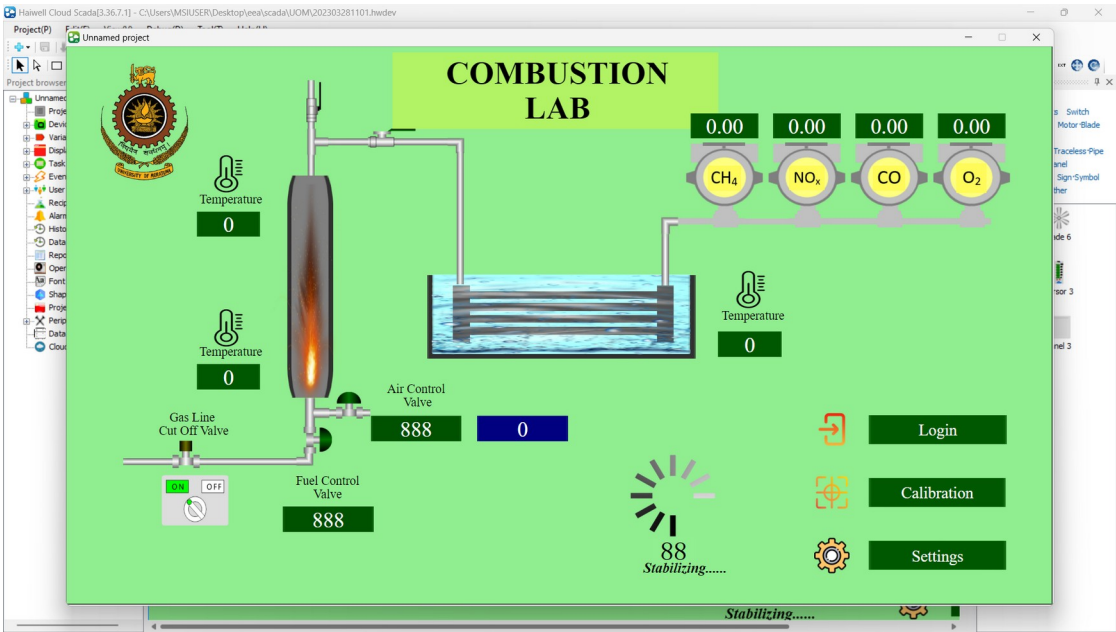
Designing the GUI for a research environment (rather than production) required understanding how researchers interpret process data during live experiments — different from a typical operator screen.

RESULTS

Enhanced SCADA GUI delivered. Research team reported improved data visibility and usability during combustion tests.

TOOLS & SOFTWARE USED	
Tool / Software	Used For
Haiwell SCADA	GUI design and process monitoring
Haiwell Happy	PLC programming
Multimeter	Sensor verification

IMAGES & TECHNICAL DRAWINGS



Haiwell SCADA user interface — customised GUI



Moratuwa University combustion chamber research rig



Combustion chamber project — instrumentation setup